

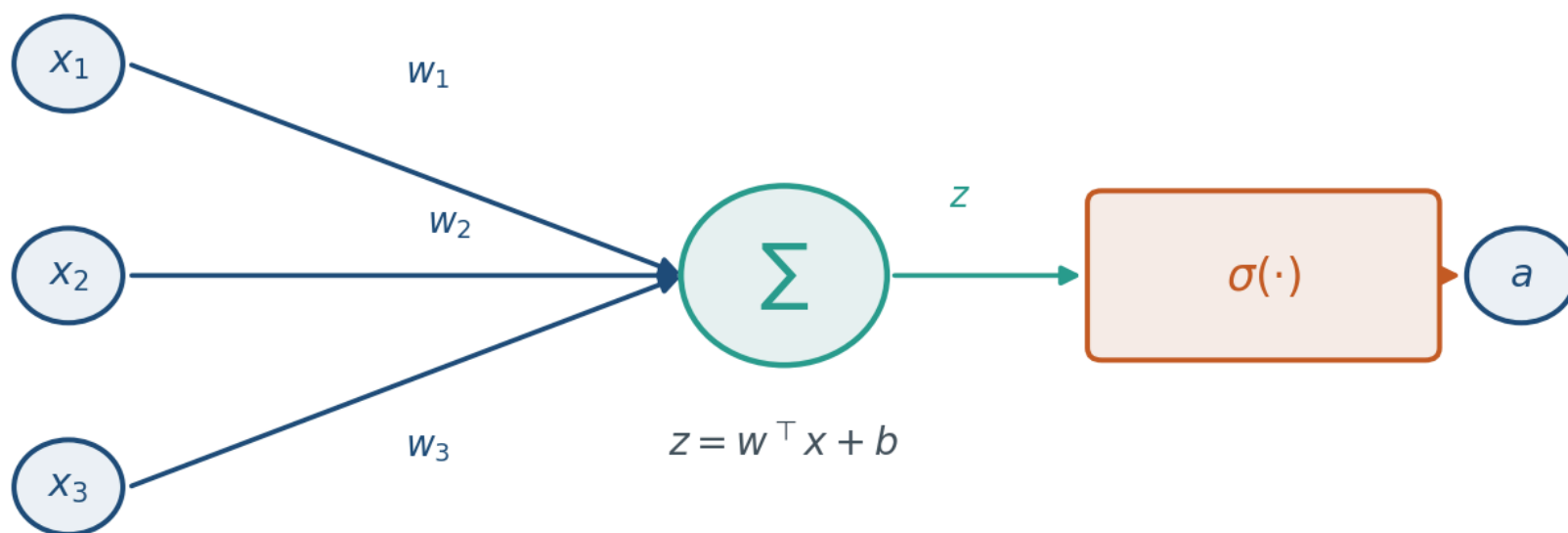
Learning goals for this block

- Write $z = w^\top x + b$ then $a = \sigma(z)$ as two boxes.
- State sigmoid, ReLU, ReLU derivative, and softmax.
- Show that two affine maps collapse to one.
- Fill Goodfellow's two-ReLU XOR table; count 3 vs 33 parameters.

One neuron

- Pre-activation $z = w^\top x + b$, then $a = \sigma(z)$.
- w and b are the parameters you train.
- If you skip z , you cannot take derivatives next note.

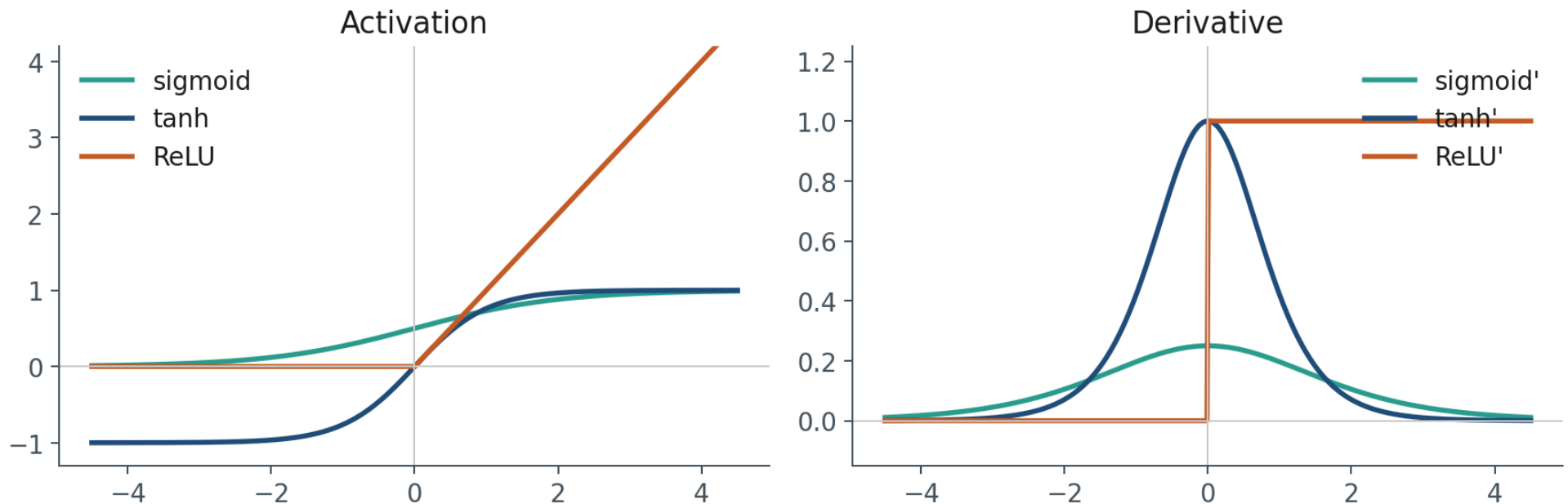
One neuron: weighted sum, then a nonlinearity



Four formulas, no Jacobian

- Sigmoid: $\sigma(z) = 1 / (1 + e^{-z})$.
- ReLU: $\max(0, z)$. Derivative: $\text{ReLU}'(z) = 1_{z > 0}$.
- Softmax: $\text{softmax}(z)_i = e^{z_i} / \sum_j e^{z_j}$.
- ReLU in hidden layers; softmax when classes compete. GELU waits.

Without a nonlinearity, stacked layers collapse to one linear map



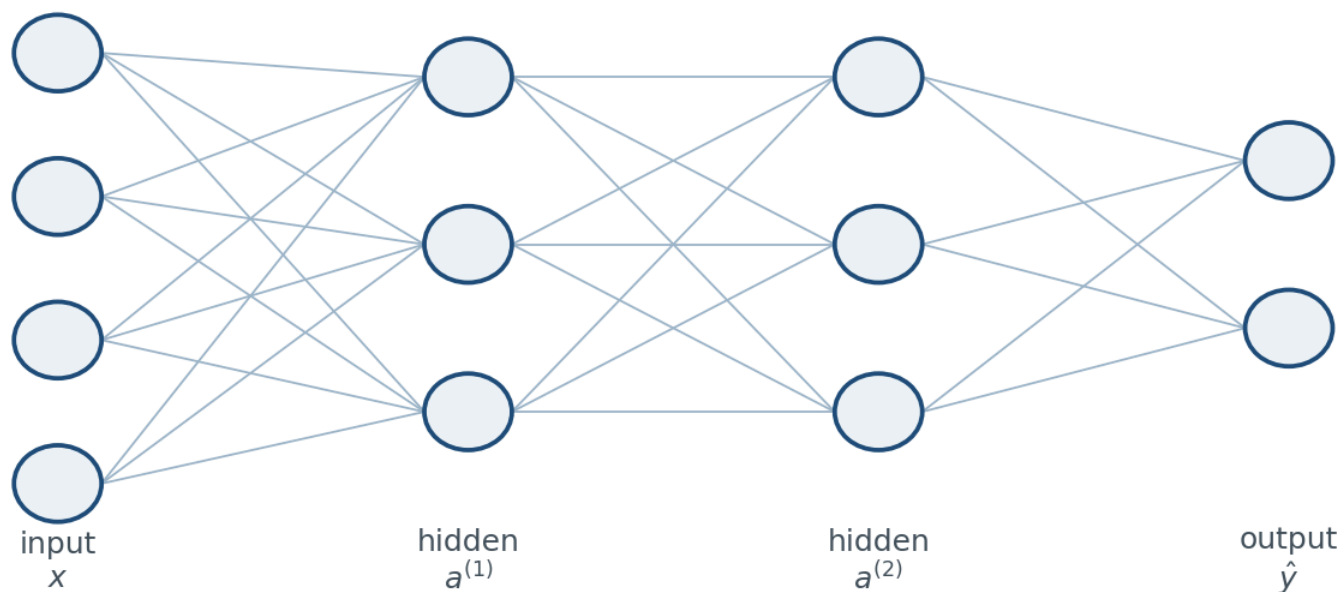
The nonlinearity is the point

- $W_2(W_1x + b_1) + b_2 = (W_2W_1)x + (W_2b_1 + b_2)$.
- Without σ , depth does not add power.
- Sigmoid saturates. ReLU is cheap and sparse (and can die).

A feedforward net

- Layer ℓ : $a^{(\ell)} = \sigma(W^{(\ell)}a^{(\ell-1)} + b^{(\ell)})$.
- $a^{(0)} = x$. Every arrow is one weight.
- A transformer block still contains this MLP, per position.

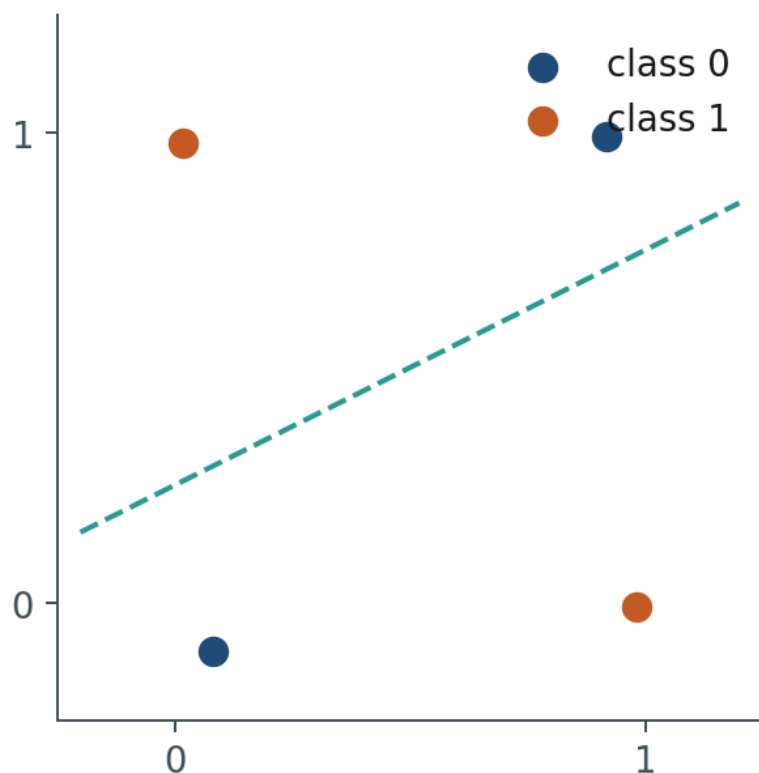
A feedforward net: each layer is $a^\ell = \sigma(W^\ell a^{\ell-1} + b^\ell)$



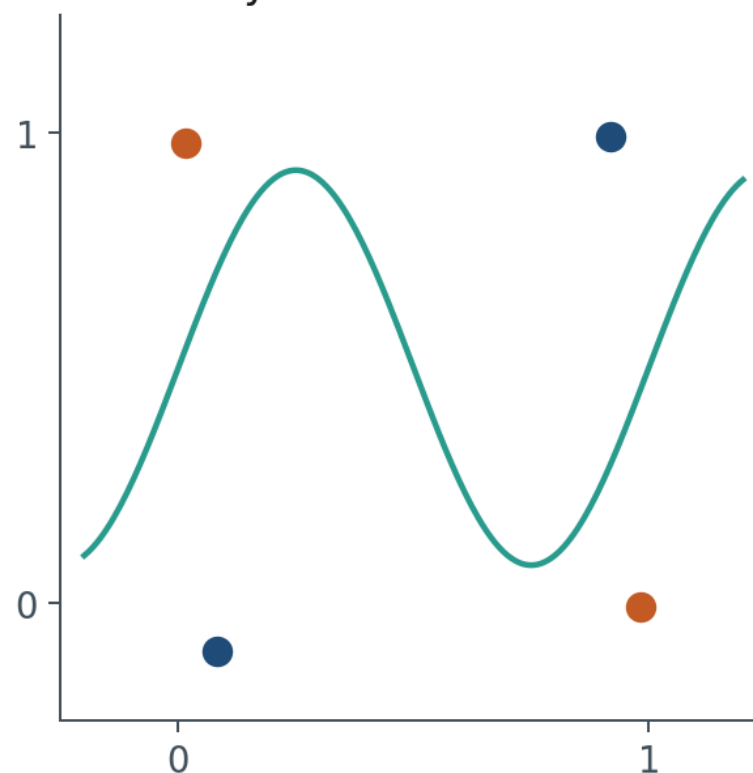
XOR: why a hidden layer exists

- One linear unit cannot separate XOR.
- A small MLP can fold the space so the classes split.
- Language models stack many such nonlinear maps.

One linear cut cannot do XOR



A hidden layer can bend the boundary



Goodfellow §6.1: an explicit XOR net

- $a_1 = \text{ReLU}(x_1 + x_2)$ (OR).
- $a_2 = \text{ReLU}(x_1 + x_2 - 1)$ (AND).
- $\hat{y} = a_1 - 2a_2$ yields 0, 1, 1, 0.

XOR with two ReLUs: OR, AND, then subtract (Goodfellow 6.1)

(x_1, x_2)	$a_1 = \text{ReLU}(x_1 + x_2)$	$a_2 = \text{ReLU}(x_1 + x_2 - 1)$	$\hat{y} = a_1 - 2a_2$
(0, 0)	0	0	0
(0, 1)	1	0	1
(1, 0)	1	0	1
(1, 1)	2	1	0

Lab 1 parameter count

- Linear classifier $\text{Linear}(2, 1)$: $2 + 1 = 3$ parameters.
- MLP $\text{Linear}(2, 8) \rightarrow \text{ReLU} \rightarrow \text{Linear}(8, 1)$: 33 parameters.
- The table already solves XOR. Extra units make training easier, not magic.
- Classical name: multi-layer perceptron. Modern name: the FFN in a block.